Question			Answer	Marks	AO	Guidance	
7	(a)		$x = 14t$	B1	1.1b	must be $x = \ldots$ or seen as the first component of the vector. Do not award for an expression that adds a vector to a scalar	SC1 for $\begin{pmatrix} 7t - 4.9t^2 + 5 \\ 14t \end{pmatrix}$ or $\begin{pmatrix} 14t - 4.9t^2 + 5 \\ 7t \end{pmatrix}$
			$y = 7t - \frac{1}{2}gt^2 + 5$	M1	1.1b	allow without +5, or if -5 seen Do not award for an expression that adds a vector to a scalar	
			So the position vector is $\begin{pmatrix} 14t \\ 7t - 4.9t^2 + 5 \end{pmatrix}$	A1	2.5	Must be a single vector. Accept $\frac{1}{2}g$ in final answer Accept $14t\mathbf{i} + \left(7t - 4.9t^2 + 5\right)\mathbf{j}$	
`				[3]			
7	(b)		Lands when $y = 0$ $7t - 4.9t^2 + 5 = 0$	M1	3.1b	Award for correct quadratic or an attempt to find value of t when their quadratic $y = 0$	
			$t = 1.95$	A1	1.1b	cao	
			gives $x = 14t = 27.3$ m	B1	1.1b	FT their t and their linear expression for x ISW where candidates find the distance from the point of projection	
				[3]			